Modular arithmetic

Definition (Modulo). If $a, b \in \mathbb{Z}$ have the same remainder after division by m, $m \mid (a-b)$, we say a and b are congruent modulo m, and write

$$a \equiv b \pmod{m}$$

Congruence modulo behaves more like an equivalence relation

Theorem. $a \equiv a \pmod{n}$ for all a (so congruence modulo n is reflexive)

if $a \equiv b \pmod{n}$, then $b \equiv a \pmod{n}$ (symmetry)

if $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $a \equiv c \pmod{n}$ (transitivity)

Theorem. Suppose that $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$. Then

$$(a \pm c) \equiv (b \pm d) \pmod{n}$$
$$ac \equiv bd \pmod{n}$$

Theorem. Let $n \geq 1$ and $a \in \mathbb{Z}$. There are unique $q \in \mathbb{Z}$ and $r \in \{0, 1, \ldots, n-1\}$ with $a = \underbrace{q}_{\text{quotient}} n + \underbrace{r}_{\text{remainder}}$.

$[c]_n$ – Congruence class of c modulo n

all values of x for which $x \equiv 1 \pmod{2}$

e.g., $[1]_2 = \{x \in \mathbb{Z} : \frac{x-1}{2} \in \mathbb{Z}\} = \{x : x \text{ is Odd}\}$

$[a]_n + [b]_n = [a+b]_n$, $[a]_n - [b]_n = [a-b]_n$, $[a]_n [b]_n = [ab]_n$

$\mathbb{Z}_n = \{[0]_n, [1]_n, \ldots, [n-1]_n\}$ is a set where we have well defined operations of addition, subtraction and multiplication, and is called the integer modulo n.

$[a]_n + [0]_n = [a]_n$ and $[a]_n [1]_n = [a]_n$ Identity

$[a]_n + [-a]_n = [0]_n$ and $[a]_n [b]_n = [1]_n$ Inverse